The galactic acceleration scale as the Jeans rate of the vacuum

Mike One · research note · 24 October 2026

The finding

The galactic acceleration scale (a_0, called g† in the paper) has been known for about 40 years to lie close to $cH_0/2\pi$. Nobody has an accepted reason why. This note proposes one, and tests it.

Proposal. The vacuum is a physical medium. Any self-gravitating medium of density ρ has a natural gravitational rate, the Jeans frequency ω_J = √(4πGρ). Applied to the vacuum's energy density ρ_Λ:

ω_J = √(4πGρ_Λ) = c√(Λ/2)

g† = c·ω_J / 2π = c√(4πGρ_Λ)/2π = 1.056 × 10⁻¹⁰ m/s²

Mechanism. Matter relaxes the vacuum wherever matter's own gravitational rate, g/c, exceeds the vacuum's Jeans rate. The transition sits where the two rates are equal.

Why it looks like cH_0 today. ρ_Λ dominates the present expansion, so the vacuum's clock and the expansion clock agree today to 1.4%. In the past H was larger, but ρ_Λ was not. So this scale stays constant while cH(z)/2π grows.

The derivation (complete chain, one postulate)

Postulate (the Law of the Dark stated as physics). The relaxed vacuum is the vacuum in its ground state. The unrelaxed vacuum is thermally excited by the vacuum's own bath. Matter's pull sets a vacuum mode; the vacuum sets the bath.

1. **Bath.** The vacuum medium has a Jeans rate ω_J = √(4πGρ_Λ), and its horizon-type temperature is kT = ħω_J/2π, the same form as the Gibbons–Hawking temperature. The 2π belongs to the bath.
2. **Statistics, which follow automatically.** For a bosonic mode in a thermal bath, P(ground state) = 1 − e^(−x). With the postulate, the relaxed fraction is r = 1 − e^(−x), and the paper's field equation ν = 1/r gives ν − 1 = ⟨n⟩ = $1/(e^x - 1)$.
3. **Symmetry, which fixes the square root.** Far from matter the unrelaxed vacuum is invariant under shifts along S, that is under rescaling (t, r) → λ(t, r). Writing x = (g_N/g†)^p, the deep regime gives g ≈ g_N^(1−p) g†^p. Scale invariance (g_N → g_N/λ², g → g/λ) forces p = ½. Equivalently, the mode energy is the geometric mean of matter's rate g_N/c and the bath frequency, which is the midpoint on S.
4. **Result, with no free numbers:** ν = 1/(1 − e^(−√(g_N/g†))) with g† = c√(4πGρ_Λ)/2π. This is the empirical radial acceleration relation (McGaugh et al. 2016) and its scale.

Relation to prior work

Milgrom (1999, “The modified dynamics as a vacuum effect”) ties MOND inertia to the Unruh temperature of an accelerated observer in de Sitter space, using temperature differences T(a) − T_Λ. - He gets $a_0 = 2(\Lambda/3)^{(1/2)}$, or $(\Lambda/3)^{(1/2)}$ in an alternative version; the two are not reconciled. - His interpolating functions are not the observed RAR form. - He leaves open which cosmological parameter sets a_0, and gives no mechanism. - No occupation numbers, ground-state probability, Jeans rate or $4\pi G\rho_\Lambda$ appear.

Differences in this work: - occupation statistics, not temperature differences - relaxed = ground state - the exact RAR function derived - the Jeans factor $\sqrt{(\Lambda/2)}$ instead of $\sqrt{(\Lambda/3)}$ - tests on four datasets in which the Λ/3 form fails

A broader literature search is still advised.

Supporting structure (derived along the way)

1. **Exact.** The galaxy law $\nu(y) = 1/(1 - e^{(-\sqrt{y})})$ is a Bose–Einstein occupation: $\nu - 1 = 1/(e^x - 1)$, with $x = \sqrt{(g_N/g\dagger)}$. Its field action is the Newtonian term plus $4\int_0^{\sqrt{y}} u^3/(e^u - 1)\,du$, which is a Planck (blackbody) integral cut at $\sqrt{y}$.
2. **Derived.** At the midpoint on S between any mass’s black-hole radius and the vacuum horizon, the Newtonian pull is $c \cdot H_\Lambda/2$, independent of the mass. This is why a single universal galaxy scale exists.
3. **Equivalent statement.** $\sqrt{(\Lambda/2)}$ is the expansion rate at the moment the vacuum is twice the matter. That is the moment of the complete sign flip, $q = -½$, the mirror of the matter era’s $q = +½$.

Tests

All rules were fixed before running. The model is $g = g_{bar}/(1 - e^{(-\sqrt{(g_{bar}/g\dagger)})})$ with no parameter fitted to the data being tested.

Dataset	Jeans (constant)	Expansion cH(z)/2π	Standard Λ/3 form
SPARC, 153 rotation curves, z ≈ 0 (paper’s own likelihood, ΔBIC vs free fit)	**−7.8**, best of 100 candidates	−7.2	+53, fails
Mistele+2024 lensing circular velocities, z ≈ 0.3 ($\Delta\chi^2$ vs best fit)	**+3.2**, passes	+0.9, passes	+19.0, fails
KiDS-1000 lensing, late-type and blue only, z ≈ 0.25 ($\Delta\chi^2$ vs best fit)	**+1.2**, passes	+0.2, passes	+13.8, fails
High-z kinematics, 145 galaxies, z = 0.7–2.5 (paper’s redshift test)	consistent (1.7σ)	**excluded (3.4σ)**	consistent

Result. The Jeans formula is the only candidate consistent with all four datasets. The standard $\Lambda/3$ form fails three independent datasets.

Search history (for honesty)

- 100 candidate formulas were tested on SPARC. None passed all three rounds: galaxies, redshift, and a physical mechanism.
- Tested and ruled out:
  - the plain vacuum horizon temperature
  - the expansion rate
  - the rate at which the expansion is slowing
  - the deceleration parameter
  - the changing-horizon (Hayward–Kodama) temperature
  - the conformal coupling $\xi = 1/6$
  - the standard de Sitter numbers ($\sqrt{2}$, 3/2, 2)
  - several Bose–Einstein counting factors
- Tried as derivations of the factor 2, and closed:
  - the brane junction equation
  - Milgrom's temperature mechanism
  - the paper's own S(t) field, whose scaling solution needs $\lambda = 2.12$ against the fitted $\lambda \leq 0.8$
- The Jeans reading came after this search. Its value matches the best SPARC candidate, so part of its SPARC score reflects selection. The two lensing tests were run afterwards, with rules fixed first.

Open points

- The chain rests on one postulate: relaxed vacuum = ground state, with matter setting the mode and the vacuum's Jeans horizon setting the bath. It is a physical postulate, testable only by its predictions.
- At $z < 0.5$, Jeans and expansion differ by only about 0.05 dex. The separation between them rests on the high-z kinematic test, whose four samples disagree with each other ($\chi^2 = 9.9$ for 3 degrees of freedom).
- The two lensing datasets share KiDS lenses, so they are not fully independent of each other.
- **Literature check needed.** $a_0 \sim c^2\sqrt{\Lambda}$ is a known order-of-magnitude coincidence. Whether the exact form $\sqrt{(4\pi G\rho_\Lambda)}/2\pi$, or its Jeans reading, has been stated before has not been checked thoroughly.

Decisive future tests

1. **Late-type weak lensing at $z \gtrsim 1$.** Jeans predicts a constant $g\dagger$. Expansion predicts growth by a factor of about 1.8 at $z = 1$.
2. **Independence from the Hubble tension.** The Jeans scale depends on ρ_Λ, not on H_0.
3. **Stellar mass-to-light ratios to about 0.05 dex,** which would separate Jeans (1.056) from the nearby candidates.

Files

The `scripts/` folder contains every test script and output: the SPARC tournament curve, the Jeans tests and the lensing tests. They run on public data (SPARC, and R. Monjo's lensing_RAR_KiDS repository).